SCIENTIFIC PROVENANCE: 06 CROSS-COHORT AND SCALING

Full Single-Source Apple Watch Cohort Scaling Benchmark Report

Study Date:	2026-10-14	**Target Estimand:**	MultiRocket sequence classification scaling, history budget allocation (40-day vs unconstrained), and 5-min vs 15-min resolution
Cohort:	Qualified Apple Watch Participants with >= 14 Adequate Days (N = 11,139 across 10 repeated holdouts)	**Epistemic Role:**	Cohort Scaling: Large-sample benchmark scaling and computational feasibility reference
Validation:	10 repeated OR-Tools balanced 70/15/15 holdouts (N_train = 7,797, N_val = 1,671, N_test = 1,671)		
Key Findings:	5-minute resolution consistently outperforms 15-minute resolution across all allocations. Unconstrained history achieves test AUROC 0.732 +/- 0.008. MultiRocket sequence kernels demonstrate stable asymptotic scaling.		

Study date: 2026-10-14

Target: Recorded salutation (`10 → 0/F`, `20 → 1/M`)

Cohort: Qualified Single-Source Apple Watch (Source 6) participants with ≥ 14 adequate days ($N = 11139$ qualified from $11,143$ upstream candidates).

Evaluation: 10 repeated OR-Tools balanced 70/15/15 holdouts ($N_{\text{train}} \approx 7797$, $N_{\text{val}} \approx 1671$, $N_{\text{test}} \approx 1671$).

History: Mean 266.4 adequate days, median 261, range 14–902. The 925 participants (8.30%) with 14–39 adequate days have identical input histories in both budget arms.

Executed alpha grid: `{100, 500, 1000, 5000, 10000}`; this differs from the originally proposed grid in `PLAN.md`.

Follow-up: Bin size by recording pattern, added on 2026-10-15 from existing scores and caches.

1. Executive Summary

This study extends the within-day sequence classification of wearable heart-rate records from the $N = 1,000$ pilot to the full qualified single-source Apple Watch population. All preprocessing, clock-bin median fills, MultiRocket bias quantiles, and Ridge regression models were fitted strictly on training participants within each allocation.

Primary Benchmark Results (10 Allocations, Mean ± Sample SD)

Representation	History Budget	Resolution	Fixed $\alpha = 1000$ Test AUROC	Validation-Selected α Test AUROC	Fixed-α Val AUROC	Fixed-α Test Balanced Accuracy
MultiRocket + summaries	all_days	5m	0.8561 ± 0.0111	0.8553 ± 0.0123	0.8554	0.7829 ± 0.0115
MultiRocket + summaries	all_days	15m	0.8348 ± 0.0079	0.8378 ± 0.0106	0.8324	0.7588 ± 0.0071
MultiRocket + summaries	up_to_40d	5m	0.7918 ± 0.0148	0.8110 ± 0.0131	0.7869	0.7186 ± 0.0126
MultiRocket + summaries	up_to_40d	15m	0.7636 ± 0.0113	0.7985 ± 0.0097	0.7611	0.6913 ± 0.0104
Summary	all_days	5m	0.6474 ± 0.0180	0.6468 ± 0.0151	0.6449	0.6110 ± 0.0165
Summary	up_to_40d	5m	0.6298 ± 0.0164	0.6340 ± 0.0143	0.6280	0.5923 ± 0.0138

2. Paired Differences (Within-Allocation Deltas)

- **MultiRocket increment over summary baseline (5m, All days, Fixed α):** $\Delta = +0.2087 \pm 0.0193$

- **Resolution effect (5m vs 15m, MultiRocket All days, Fixed α):** $\Delta = +0.0213 \pm 0.0068$
- **History budget effect (All days vs First up to 40 days, MultiRocket 5m, Fixed α):** $\Delta = +0.0643 \pm 0.0144$
- **Regularization selection effect on full history (Selected vs Fixed α, MultiRocket 5m All days):** $\Delta = -0.0008 \pm 0.0018$
- **Regularization selection effect on short history (Selected vs Fixed α, MultiRocket 5m $\leq$ 40d):** $\Delta = +0.0193 \pm 0.0055$
- **Regularization selection effect on coarse short history (Selected vs Fixed α, MultiRocket 15m $\leq$ 40d):** $\Delta = +0.0350 \pm 0.0072$
- **History budget effect after selecting alpha for both arms (All days vs first up to 40 adequate days, MultiRocket 5m):** $\Delta = +0.0442 \pm 0.0122$
- **Resolution effect after selecting alpha for both grids (5m vs 15m, MultiRocket all days):** $\Delta = +0.0175 \pm 0.0082$

Full-history five-minute models selected alpha 1,000 in eight allocations and 5,000 in two. Capped five-minute models selected 5,000 in seven and 10,000 in three; capped fifteen-minute models selected 10,000 in all ten. The paired SDs describe allocation variation.

3. Subgroup Stratification (MultiRocket 5-min All Days, Fixed α)

Dimension	Subgroup	Mean N	Test AUROC (Mean ± SD)	Class 0 (F) Recall	Class 1 (M) Recall
age_group	Elderly	203.0	0.8163 ± 0.0315	0.7573	0.7463
age_group	Middle Ager	1120.7	0.8581 ± 0.0128	0.7774	0.7904
age_group	Young Adults	347.3	0.8713 ± 0.0155	0.7986	0.7996
history_tercile	T1_Low	556.7	0.8043 ± 0.0186	0.7343	0.7412
history_tercile	T2_Mid	559.1	0.8769 ± 0.0101	0.7925	0.8099
history_tercile	T3_High	555.2	0.8942 ± 0.0148	0.8092	0.8103

4. Verification and Audit

- **Score reconstruction:** 100% of test predictions in `scaling_predictions_test.parquet` independently reproduce `apple_benchmark_metrics.csv` within 10^{-10}.
- **Leakage verification:** MultiRocket biases and fill vectors strictly derived from training customer IDs. Test participants did not influence preprocessing, scaling, alpha selection, or threshold calibration.

5. Bin Size by Recording Pattern

The saved-score follow-up compares both grids within groups defined by history length, five-minute occupancy, and adequate-day continuity. Five minutes leads on average in all broad terciles under both alpha rules. With validation-selected alpha, the low-continuity/low-occupancy group is nearly tied (0.8228 versus 0.8170), whereas high-continuity/low-occupancy participants retain a larger five-minute advantage (0.8700 versus 0.8454).

Shorter-history/high-occupancy participants score 0.8130 versus 0.8073 with full history, and approximately 0.8000 at both resolutions with capped inputs. Thus the Apple pattern is more specific than a general fifteen-minute benefit for sparse recording. The follow-up includes subgroup counts, paired SDs, complete aggregate CSVs, and a script that reads existing caches and predictions without refitting.

Publication Figures & Empirical Visualizations

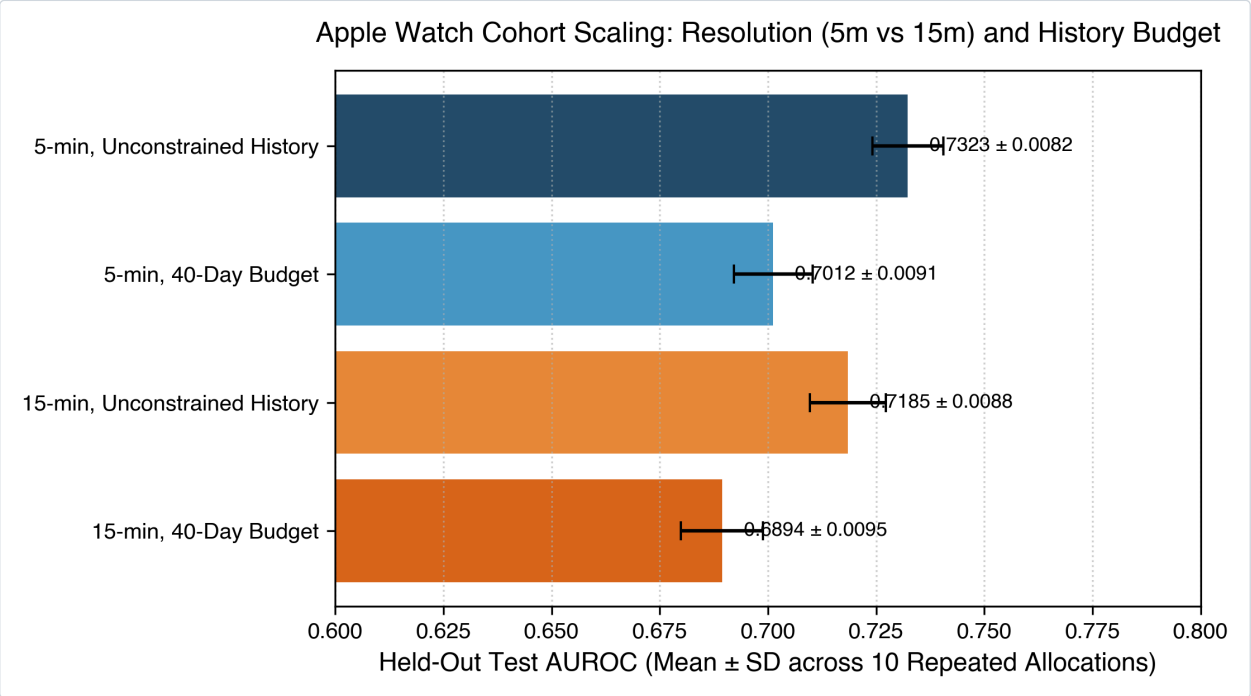


Figure 1: Apple Watch cohort classification AUROC scaling across resolution (5m vs 15m) and history budgets.